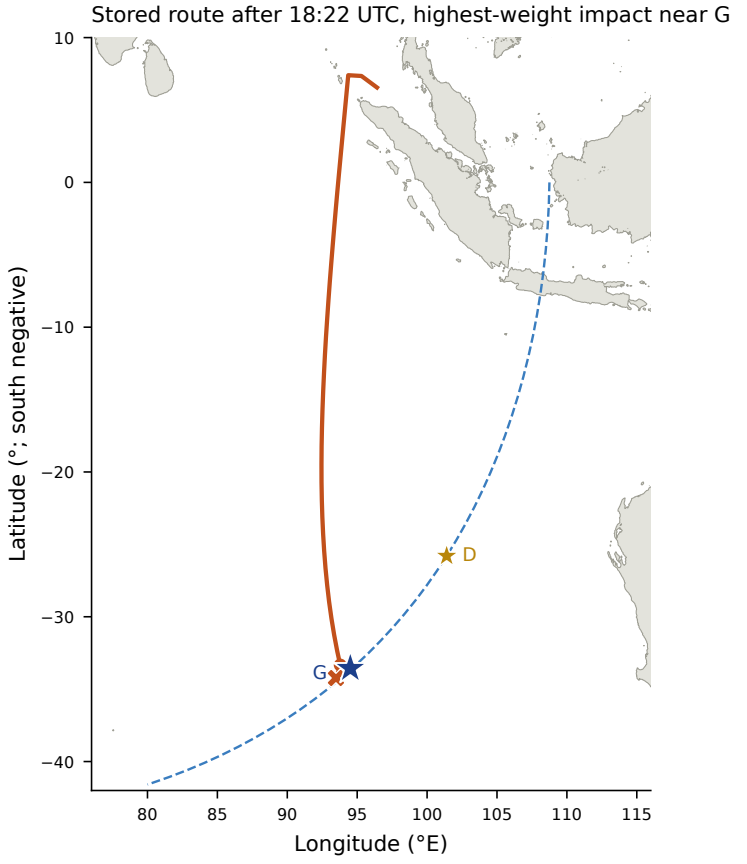
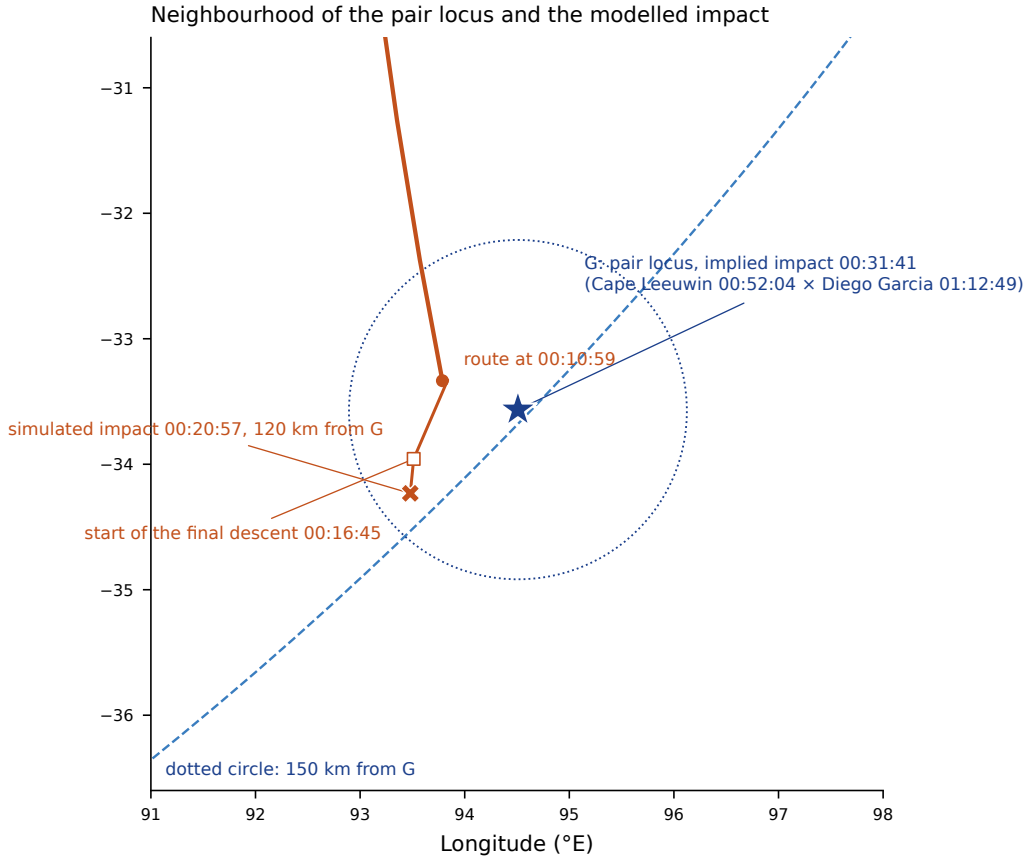
The highest-weight simulated impact within 150 km of the pair locus G hits at 00:20:57, 11 min before the time the pair implies





(i) The orange line is one stored flight path from the simulation. Its impact is the one with the highest weight within 150 km of the pair point G. It hits the sea 11 minutes before the time that the pair of sounds gives. The flight path and the impact are linked by position at 00:11, not by a direct record. The two sounds (Cape Leeuwin 00:52:04, Diego Garcia 01:12:49) have a time difference that agrees with one point on the arc, 33.6°S 94.5°E. Pairs like this occur by chance: the test gives p 0.019 from 36 tests, which is a chance level. The Diego Garcia sound is not kept when the airgun noise is removed. The sound direction that Kadri published for 00:52 does not point to this place. The impact would need about 330 times more sound than the calibration crash gives.

(ii) Pair G: Cape Leeuwin H01W 00:52:04 UTC (DIGITISED Kadri 2024 Fig. 9 panel c; envelope matched filter M1b p 0.043; Kadri's 'text time 00:52:03', 'caption time 00:54:30') x Diego Garcia South H08S 01:12:49 (DIGITISED panel e, raw, M1b p 0.039; the airgun-suppressed residual does not keep it, p_B 0.30); lag 1,245 s inside the predicted 809-1,536 s; 19 lag-consistent pairs vs 8.8 expected, permutation p 0.019 over 36 receiver-pair tests (chance level: a smallest p this low arises about half the time). Arc locus 33.6°S 94.5°E, implied impact 00:31:41; eta for P_D(both) = 0.5 is 0.069 = 330x the F-35A anchor 2.1e-4. BEARING: the locus lies at 269.0° from H01W (1,812 km) and 144.8° from H08S (3,657 km), WGS84; Kadri gives 57° (text p. 9) or 306.18° (Table 1 caption) for the 00:52 signal; neither is within 37° of the locus bearing; nearest Table 1 row 00:52:50 at 274.84° (46 s later). Trajectory: core run J waypoints stratum, seed 11, stored route sample 1392 (routes.npy, lat/lon every 600 s from 18:01:49; drawn from 18:21:49 to 00:01:49 and interpolated to 00:10:59); joined to end-of-flight row (seed 11, parent hand-off row 4444) by POSITION at 00:10:59 (route-to-parent distance 4.8 km): the stored routes carry no row identity, so the join is an association, not a link. End-of-flight segment: hand-off position 00:10:59 -> start of final descent (takeover, 00:16:45) -> impact 00:20:57 at 34.23°S 93.48°E; family B (descent begun before fuel exhaustion); max descent rate 40,722 ft/min; share of the D0' p 0.5 waypoints-stratum mass 2.7e-05 (mass within 150 km of G 0.045). Weights: run_j_windows.ln_weights(D0') x p_factor(0.5), seeds pooled equally (24 seeds). Source posterior run J, 00:19 Held Out (D0'), dive probability p 0.5; PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, +10.4/+7.4 dB per octave at H01W/H08S) and C2 (coupling efficiency not calibrated; eta swept over four decades, F-35A upper anchor). Module branch hypothesis/hydroacoustics 9b27786; searches results-data/searches_run_j.